Chapter 7 exercises

Work over $\mathbb{R}$ unless stated otherwise. State every matrix shape. For a full SVD, check $U^TU = I_m$, $V^TV = I_n$, and $A = U\Sigma V^T$. A reduced factorization uses identities of size r.

CH07-E1

Let

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 0 \end{bmatrix}.$$

1. Compute A^TA and its eigenvalues and eigenvectors.
2. Find the positive singular values and corresponding left singular vectors.
3. Give a reduced SVD.
4. Complete the left vectors to an orthonormal basis of $\mathbb{R}^3$, give a full SVD, and identify the additional vector's fundamental subspace.

CH07-E2

For the matrix in CH07-E1,

1. Give the dimensions of $N(A), N(A^T), C(A), C(A^T)$.
2. Give their bases using singular vectors.
3. Check $Av_i = \sigma_i u_i$.
4. Verify the orthogonal decompositions of $\mathbb{R}^2$ and $\mathbb{R}^3$.

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CH07-E3

Let

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}.$$

1. Find rank, singular values, and a reduced SVD.
2. Reconstruct the matrix from $\sigma_1 u_1 v_1^T$.
3. Compute $\|A\|_2$, $\|A\|_F$, A^+.
4. For $b = (1, 3)^T$, find all least-squares solutions and the minimum-norm solution.

CH07-E4

> Let
> $A = \begin{bmatrix} 4 & 0 \\ 3 & 5 \end{bmatrix}$.
> Although A is not symmetric, explain why it has an SVD. Without computing a full eigendecomposition, prove invertibility and $\kappa_2(A) \geq 1$ using the singular-value structure. If a perturbation makes one positive singular value zero, describe the change in rank and conditioning.

CH07-E5

> A matrix has singular values $9, 4, 1, 0$.
>
> 1. Find its rank.
> 2. For $k = 1, 2, 3$, give the best rank-k errors in spectral and Frobenius norms.
> 3. State how to compute the retained energy fraction.
> 4. Find the smallest k permitting spectral error at most 4.

CH07-E6

> Three two-dimensional samples are the columns of
> $X_0 = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}$.
>
> 1. Verify that the matrix is centered.
> 2. Find both principal directions and score rows.
> 3. Compute sample covariance and the largest variance contribution.
> 4. Describe the first direction after multiplying row 2 by 100. Explain the resulting standardization decision.
> 5. Retain only the first principal direction. Compute every projection and residual, then check their squared total against the discarded singular value.
> 6. Add $\mu = (2, 3)^T$ to every sample. What changes in the centered principal directions, the best line in the original coordinates, and the minimum error?

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CH07-E7

Given

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

1. Find a reduced SVD.
2. Compute $A^{+}b$ and $AA^{+}b$.
3. Verify residual orthogonality.
4. Relate the complete minimizer set to the minimum-norm solution.

CH07-E8

Let square $A \in \mathbb{R}^{n \times n}$ have polar decomposition $A = QS$.

1. Derive $Q = UV^{T}$ and $S = V\Sigma V^{T}$ from SVD.
2. Prove that Q is orthogonal and S symmetric positive semidefinite.
3. If A is invertible, prove that S is positive definite and invertible.
4. Relate an orientation-reversing Q to the sign of $\det A$.

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CH07-E9

1. Prove that $AA^{+} = U_r U_r^{T}$ and $A^{+}A = V_r V_r^{T}$ are orthogonal projectors.
2. Prove $AA^{+}A = A$ and $A^{+}AA^{+} = A^{+}$.
3. Explain why truncation through a tied singular value can give nonunique retained subspaces.
4. Explain why flipping only u_1, without flipping v_1, changes reconstruction.

CH07-E10

1. Use the optimal spectral-norm low-rank approximation formula to prove $|\sigma_i(A) - \sigma_i(B)| \le \|A - B\|_2$.
2. For $A = \mathrm{diag}(1, 10^{-8}, 10^{-12})$ and $\|A - B\|_2 \le 10^{-10}$, find a guaranteed rank lower bound and give compatible rank-2 and rank-3 models. Is the true rank determined?
3. For $R = \begin{bmatrix} 1 & 10^6 \\ 0 & 1 \end{bmatrix}$, prove $\sigma_{\min} \le 10^{-6}$ without a cancellation-prone eigenvalue formula. Construct a rank-1 matrix within 10^{-6}. What does this show about testing only QR diagonal entries?
4. For fixed absolute threshold t and perturbation radius δ, give sufficient conditions for classification above or below the threshold. Explain the sensitivity of comparing only σ_i with t.

CH07-E11

Use $V = \frac{1}{5}\begin{bmatrix} 4 & -3 \\ 3 & 4 \end{bmatrix}$, $U = \frac{1}{5}\begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix}$, $\Sigma = \mathrm{diag}(3, 2)$ and x=(1,0)ᵀ. Calculate each of Vᵀx, ΣVᵀx and UΣVᵀx; check by direct multiplication with A=UΣVᵀ. List the norms at all four stages. Prove the output-length bounds for every unit input and characterize both equality cases. Why is the output length for this x not a singular value?

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CH07-E12

1. Let $A = \operatorname{diag}(8, 3, 2)$. Allow approximations of rank at most one; all quantities are real. Write truncated A_1 and error E. Derive $\|E\|_2$ directly from unit inputs and compute $\|E\|_F$. Why does $\|E(e_2 + e_3)\| = \sqrt{13}$ not contradict the spectral value? Normalize the input.
2. Within $B_c = \operatorname{diag}(c, 0, 0)$, find all c attaining the global spectral minimum and all c attaining the global Frobenius minimum. Explain why this does not list every possible spectral minimizer.
3. For $B = \begin{bmatrix} 8 & 8 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, find a unit kernel vector in span(e_1,e_2) and verify the spectral lower-bound argument without simultaneous diagonalization.
4. Use projection capacities α_i and columnwise Pythagoras to prove global uniqueness of the Frobenius minimizer, not merely uniqueness within the Bc family.
5. Replace A by $D = \operatorname{diag}(3, 3, 1)$. Set $w_\theta = (\cos\theta, \sin\theta, 0)^T$, $C_\theta = 3w_\theta w_\theta^T$. Compute both errors and prove these are different optimal rank-one matrices. Distinguish the fixed complete tied singular subspace from its selected retained direction.

CH07-E13

Let $A = [1\ 1]$, $B = [1\ 0]^T$. Find their ranks, $(AB)^+$, and B^+A^+. Explain why maximum possible rank for both factors does not justify applying the reversed inverse rule for invertible square matrices.

References and attribution

Lin Cing, handwritten linear algebra notes. Gilbert Strang, Introduction to Linear Algebra, 5th ed., Chapter 7.

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